

Multiple Choice

1. **Answer:** D

Options: A) Na_2CO_3 ; B) Na_3PO_4 ; C) Na_2S ; D) NaCl

Note: Correct option: D - NaCl

2. **Answer:** B

Options: A) 23.56 GHz; B) 42.58 MHz; C) 74.34 kHz; D) 13.93 MHz

Note: Correct option: B - 42.58 MHz

3. **Answer:** A

Options: A) 0.050 M AlCl_3 ; B) 0.100 M NaCl ; C) 0.050 M CaCl_2 ; D) 0.100 M HCl

Note: Correct option: A - 0.050 M AlCl_3

4. **Answer:** A

Options: A) MgO ; B) MgS ; C) NaF ; D) NaCl

Note: Correct option: A - MgO

5. **Answer:** A

Options: A) hexane; B) 2-methylpentane; C) 3-methylpentane; D) 2,3-dimethylbutane

Note: Correct option: A - hexane

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' H_3O^+ / H_2O '.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It has little effect.'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '5'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Ultraviolet. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** T_2 , unlike T_1 , is sensitive to molecular motions at the Larmor frequency.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.